

scripts/run-challenge-matrix.sh

— User Guide

Last verified: 2026-05-16 (HelixQA Option 1 wiring landed — `--include-helixqa` flag added) **Inheritance:** HelixConstitution §11.4.18 + Lava §6.AE.6 + §6.X (Container-Submodule Emulator Wiring) + §6.I (Multi-Emulator Container Matrix)

Overview

Operator entry point for §6.AE gate-mode Challenge Test runs. Pre-bakes the §6.AE.2 minimum AVD matrix (API 28 / 30 / 34 / latest stable × phone + tablet) and delegates to submodules/containers/cmd/emulator-matrix --runner=containerized per §6.X.

Honest pre-flight

The script DETECTS the §6.X-debt darwin/arm64 host gap (no /dev/kvm available to podman containers; macOS HVM not exposed to podman) and, when detected, REFUSES to claim a successful gate run. Instead it:

1. Validates arguments + matrix minimum
2. Writes <evidence-dir>/host-preflight.json with the host-gap classification
3. Exits 2 (NOT 0) — gate-host ineligible

Per §6.J/§6.L: no false-pass; honest unblock report. Real gate runs require a Linux x86_64 + KVM host.

Usage

Run ALL Challenges on the §6.AE.2 minimum matrix
`bash scripts/run-challenge-matrix.sh`

Run a specific Challenge
`bash scripts/run-challenge-matrix.sh \`
 `--test-class`
 `lava.app.challenges.Challenge01AppLaunchAndTrackerSelectionTest`

Add TV-class AVD (when feature touches TvActivity)
`bash scripts/run-challenge-matrix.sh --add-tv`

```
# Add foldable AVD
bash scripts/run-challenge-matrix.sh --add-foldable

# Override "latest stable" API level
bash scripts/run-challenge-matrix.sh --latest-api 36

# Skip APK rebuild
bash scripts/run-challenge-matrix.sh --no-build

# ALSO invoke the 11 HelixQA Challenge scripts (Option 1 wiring)
bash scripts/run-challenge-matrix.sh --include-helixqa
```

Inputs

Arg	Description
<code>--test-class <fqcn></code>	Specific Challenge to run (default: all under <code>lava.app.challenges</code>)
<code>--evidence-dir <dir></code>	Output directory (default: dated under <code>.lava-ci-evidence/</code>)
<code>--no-build</code>	Skip the <code>:app:assembleDebug</code> step
<code>--latest-api <N></code>	Override the "latest stable" API level (default: 36)
<code>--add-tv</code>	Add a TV-class AVD to the matrix
<code>--add-foldable</code>	Add a foldable AVD
<code>--include-helixqa</code>	ALSO invoke <code>scripts/run-helixqa-challenges.sh</code> (the 11 HelixQA Challenge scripts per Option 1 wiring). OFF by default so existing matrix runs are unaffected. HelixQA runs on the HOST BEFORE the AVD matrix → independent of §6.X-debt darwin/arm64 gate-host gap. HelixQA wrapper evidence lands at <code><evidence-dir>/helixqa/</code> . Non-zero HelixQA exit promotes the final aggregate exit code (matrix exit dominates if both fail). See <code>docs/scripts/run-helixqa-challenges.sh.md</code> for full details.

§6.AE.2 mandatory minimum matrix

```
CZ_API28_Phone:28:phone
CZ_API30_Phone:30:phone
CZ_API34_Phone:34:phone
CZ_API34_Tablet:34:tablet
CZ_API<latest>_Phone:<latest>:phone
```

Plus TV / foldable when explicitly requested by `--add-tv` / `--add-foldable`.

Sub-minimums are permitted for development iteration; the gate row's gating: true flag is only set when the full minimum is satisfied + every config dimension (theme/locale/density per §6.AE.2) is covered.

Outputs

- <evidence-dir>/host-preflight.json — host-gap classification (always written)
- <evidence-dir>/real-device-verification.{md,json} — per-AVD attestation rows (only on Linux x86_64 + KVM gate-host)
- <evidence-dir>/helixqa/helixqa-attestation.json + per-script logs — only when --include-helixqa is set

Exit codes

Code	Meaning
0	Matrix completed; all AVDs passed
1	At least one AVD failed boot OR at least one test failed (real Containers CLI exit)
2	Gate-host ineligible (darwin/arm64 OR no KVM); honest unblock-needed report written

Cross-references

- submodules/containers/cmd/emulator-matrix (the underlying runner)
- tools/lava-containers/vm-images.json (matrix manifest)
- scripts/run-emulator-tests.sh (older sister-glue; same delegation, different default arguments)
- scripts/run-helixqa-challenges.sh + docs/scripts/run-helixqa-challenges.sh.md (the HelixQA wrapper invoked by --include-helixqa)
- docs/plans/2026-05-16-helixqa-integration-design.md (Option 1 wiring design)
- .lava-ci-evidence/sixth-law-incidents/2026-05-13-emulator-container-darwin-arm64-gap.json (the standing §6.X-debt incident)
- Lava CLAUDE.md §6.AE + §6.X + §6.I